The Role of Artificial Intelligence in Streamlining Prior Authorizations within a Health-System Specialty Pharmacy

Dylan McWilliams, PharmD[1]; Anneliesa Henslee, PharmD, MS[1]; Trisha Jordan, PharmD, MS[1]; Liz Hall, PharmD[1]; Christine Prusa, PharmD, BCACP[1]; Steve Deerwester CPhT, LSW[1]; Brianne Porter, PharmD, MS[2]

1. The Ohio State University Wexner Medical Center | 2. The Ohio State University College of Pharmacy

Authors have no conflict of interests to declare

Contact Information:
Dylan McWilliams
Dylan.McWilliams@osumc.edu

THE OHIO STATE UNIVERSITY
WEXNER MEDICAL CENTER

Background

- Prior authorizations (PA) are required for most specialty medications in which can delay therapy, increase administrative burden, and negatively affect patient care.[1]
- At The Ohio State Wexner Medical Center (OSUWMC), PAs are primarily completed electronically (ePA) through CoverMyMeds by dedicated prior authorization coordinators (PACs).
- Manual processes contribute to inefficiencies, treatment delays, and increased workload for pharmacy and clinic teams.
- Artificial intelligence (AI) has emerged as a potential solution to streamline PA workflows, reduce turnaround time, and improve approval rates.[2,3]
- No published studies currently evaluate AI's impact on PA efficiency within a health-system specialty pharmacy setting.

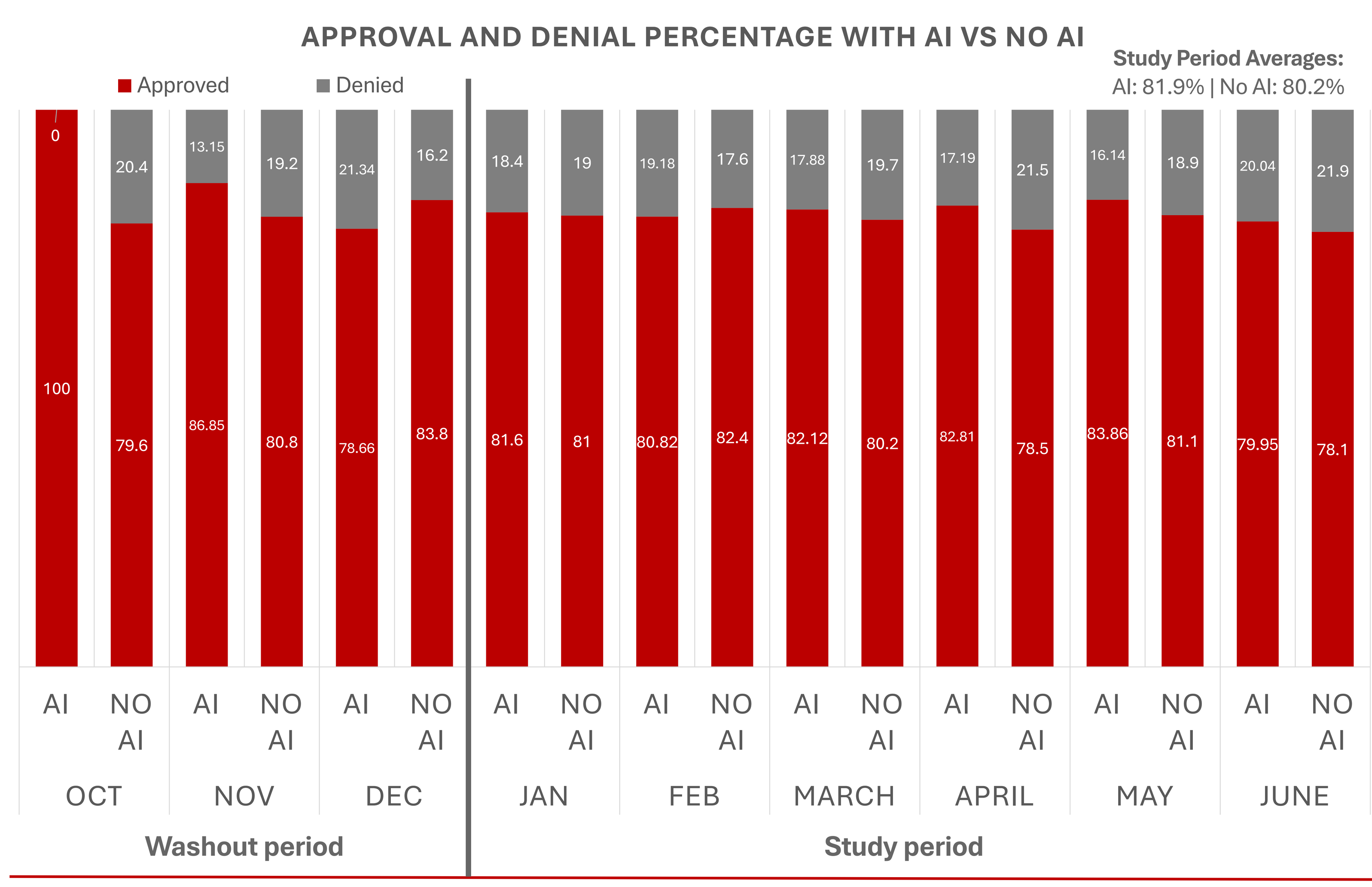
Study Objective

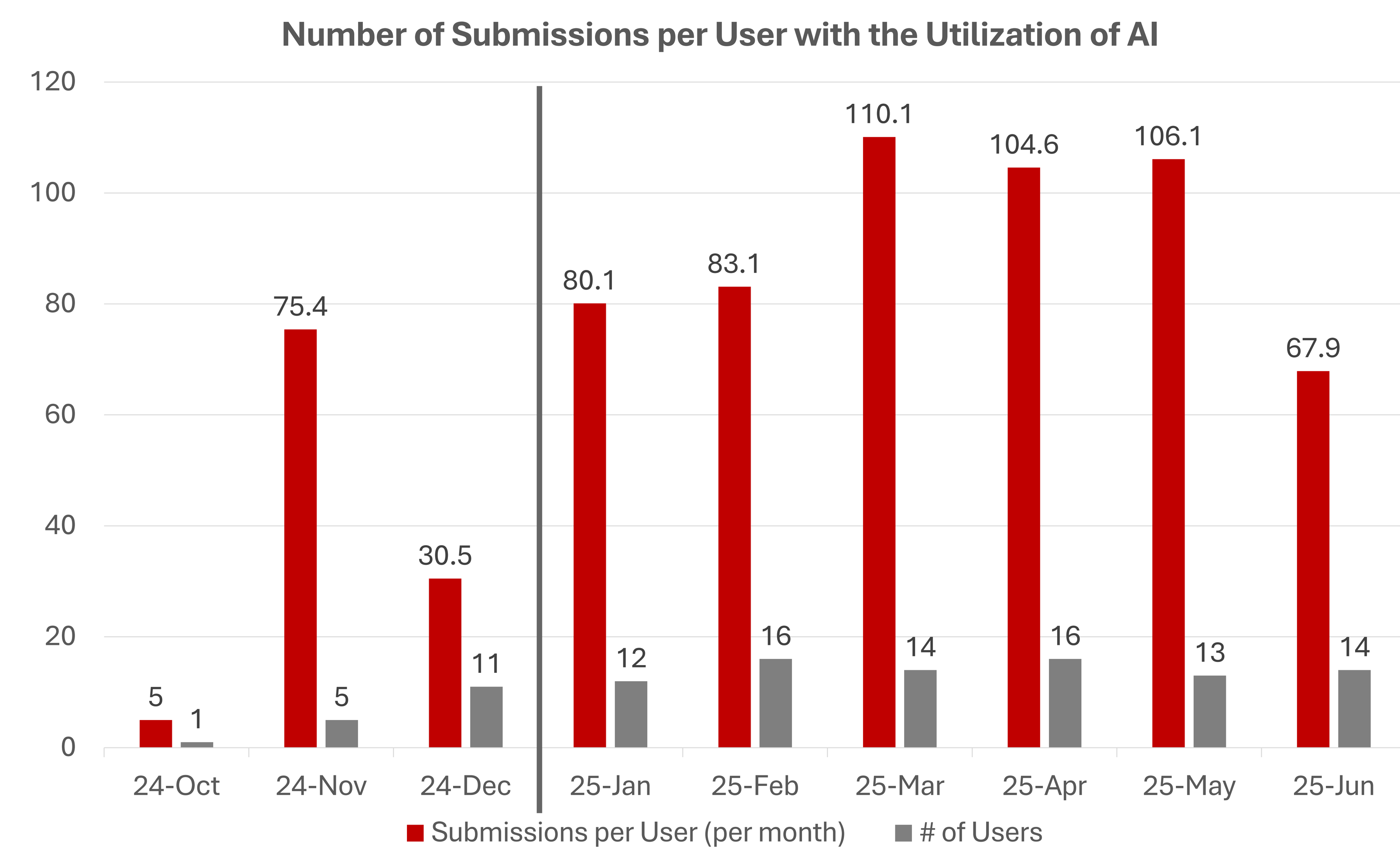
To determine the impact of artificial intelligence on prior authorization approval rates and turn-around time within a health-system specialty pharmacy.

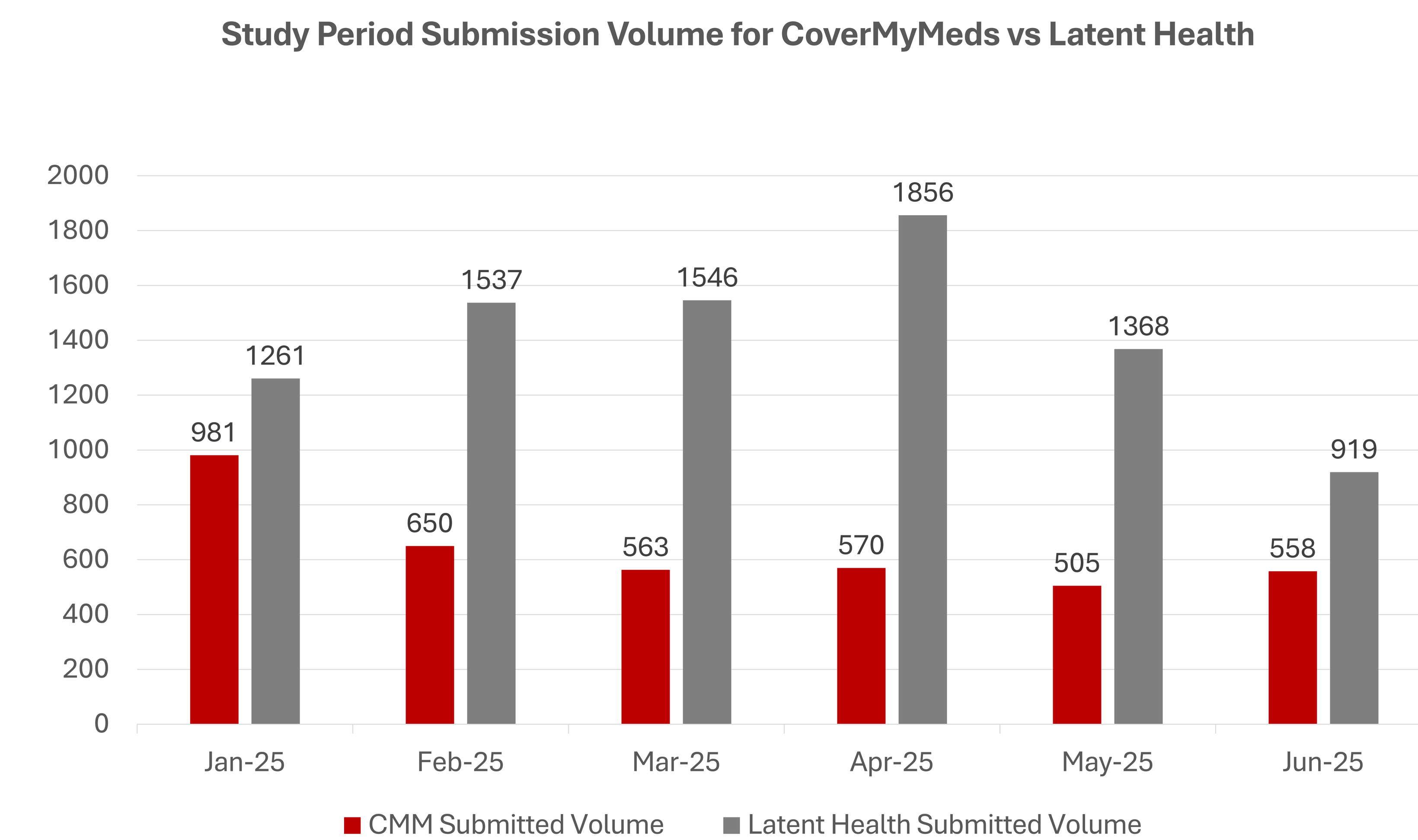
Methods

- Prospective, single-center quality improvement study evaluating all specialty pharmacy PAs from providers at The Ohio State University Wexner Medical Center between September 2023–October 2025. Pre-implementation data (Sept 2023–Dec 2024) compared to post-Latent Health implementation data (Oct 2024–Dec 2025). The washout period for the implementation of Latent Health was Oct-Dec 2024.
- Data collected included total active time on PA, clinical time, and approval/denial status from CoverMyMeds and Latent Health dashboards.
- **Inclusion:** Patients prescribed medications requiring PA by OSUWMC providers.
- **Exclusion:** Prescriptions not requiring PA.

Results







Conclusion

- Implementation of Latent Health improved PA approval rates, with an average increase from 80.2% to 81.9% across the study period.
- These findings potentially support the utility of AI as a scalable solution for health-system specialty pharmacies aiming to reduce administrative burden and optimize medication access.

References

1. Brown JD, Lawton AR, Costales B, Stottlemyer JR, Crannage AJ. Analysis of prior authorization success and timeliness at a community-based specialty care pharmacy. J Am Pharm Assoc (2003). 2021;61(3):e148-e152. doi:10.1016/j.japh.2021.01.001
2. Wright A, Sittig DF. Could an artificial intelligence approach to prior authorization be more human? J Am Med Inform Assoc. 2023;30(5):1025-1029. doi:10.1093/jamia/ocad016
3. Blanchard K, Wilkinson J, West D. Perceptions of prior authorization burden and solutions. Health Serv Res Pharm. 2023;8(1):e096. doi:10.1093/haschl/qxae096